Evaluating AI Agents

Evaluation in the Time of LLMs





DeepLearning.AI

Talking Head

In this lesson:

- LLM model evaluation vs LLM system evaluation
- Similarities and differences to software development testing
- Agent use cases and why you should evaluate agents
- Agent testing and iteration tools

Scope of evaluation

LLM Model Evaluation

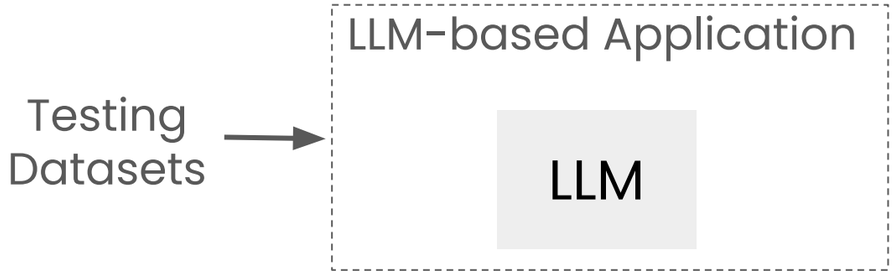
- Measure the general language understanding of the foundational models.



- Example of Benchmark Datasets:
  - MMLU (multiple-choice questions covering math, philosophy, medicine...)
  - HumanEval (code generation)

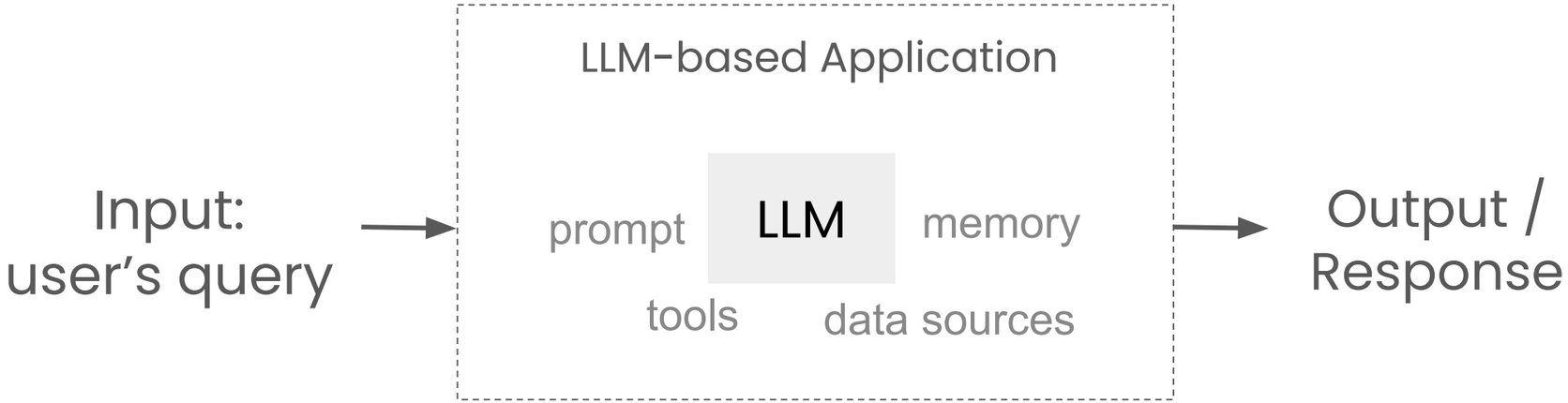
LLM System Evaluation

- Evaluate how well the entire application, including the LLM, performs (meeting business requirements).



- Testing Datasets can be manually created, synthesized, or curated from the application
  - i.e. real world data

LLM-based applications



- Examples: Agents, RAG Systems

A paradigm shift from traditional software testing





A paradigm shift from traditional software testing

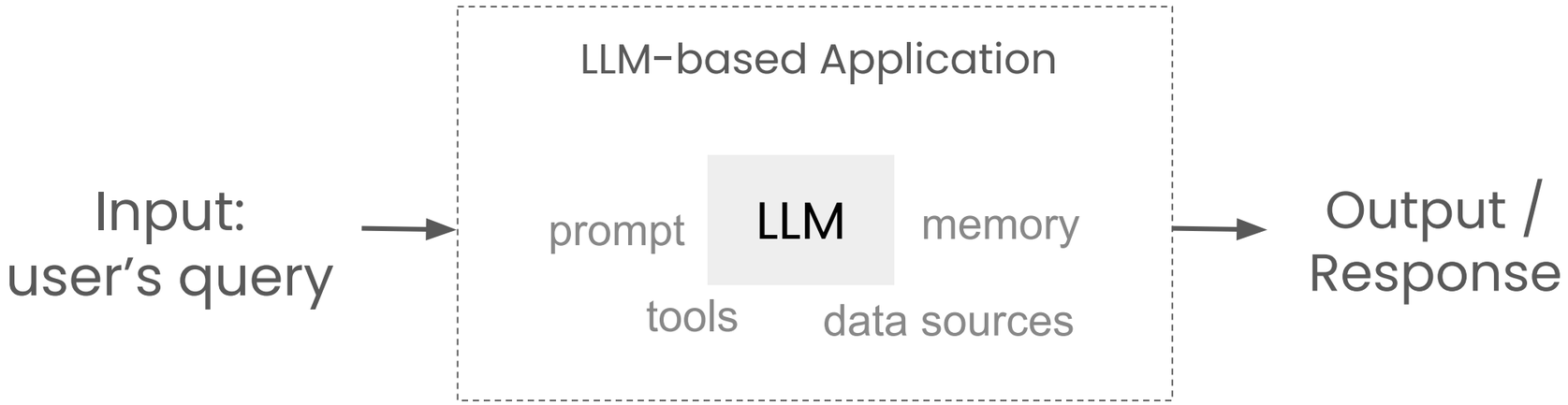
- Traditional software testing:
  - Unit testing: testing individual component of the software application
  - Integration testing: testing how different components work together

- Testing in the time of LLMs:
  - Non-deterministic nature of LLMs: outputs can vary even with the same inputs
  - Focus on testing the application's ability to respond to users' specific tasks
  - Examine the output quality (relevance, coherence)

Common Types of Evaluations for LLM Systems

- Hallucinations
- Retrieval relevance
- Q&A on retrieved data
- Toxicity
- Summarization performance
- Code writing correctness and readability

LLM-based applications: agents



- What are agents?
- What could go wrong with agent-based systems?

What are agents?

- Agents are software based systems that can take actions on behalf of a user utilizing reasoning
  - Reasoning: powered by AI (LLMs)
  - Routing: interpreting request and determining the correct tool
  - Action: executing code/tools (utilizing APIs, making LLM call)

Agent use cases

- Personal assistants
- Desktop or Browser agents
- Automated scraping and summarization
- Research assistant

arize

DeepLearning.AI

Popular Agent Frameworks

All Courses > Short Courses > AI Agents in LangGraph

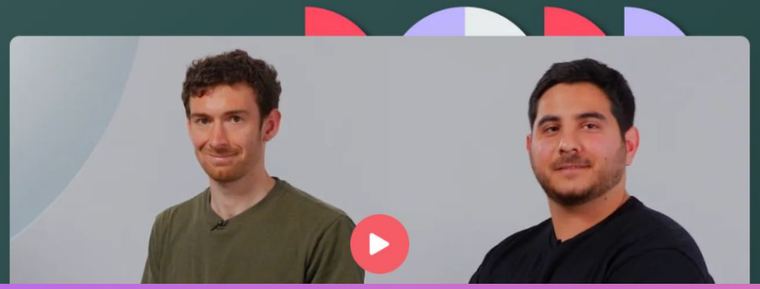
Short Course | Intermediate | 1 Hour 25 Minutes

AI Agents in LangGraph

Instructors: Harrison Chase, Rotem Weiss

LangChain Tavily

Enroll for Free



All Courses > Short Courses > Building Agentic RAG with LlamaIndex

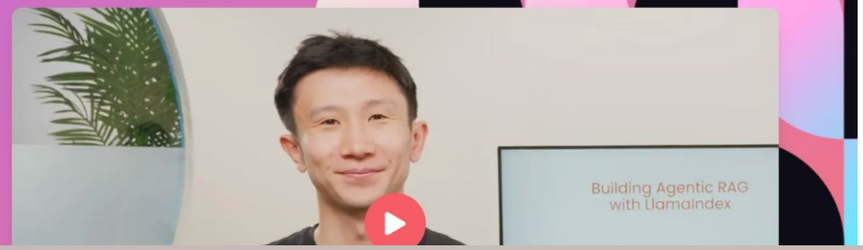
Short Course | Beginner | 46 Minutes

Building Agentic RAG with LlamaIndex

Instructor: Jerry Liu

LlamaIndex

Enroll for Free



All Courses > Short Courses >
Practical Multi AI Agents and Advanced Use Cases with crewAI

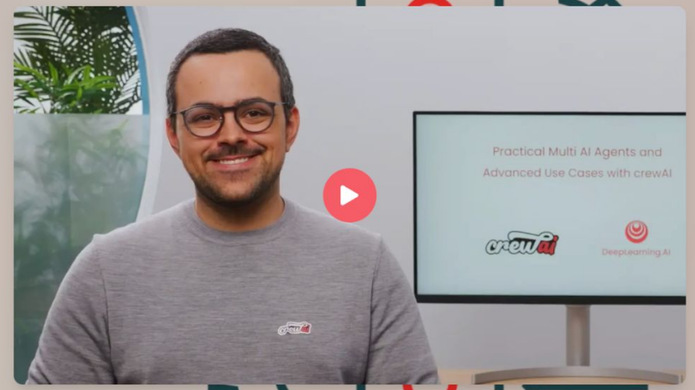
Short Course | Beginner | 2 Hours 43 Minutes

Practical Multi AI Agents and Advanced Use Cases with crewAI

Instructor: João Moura

crewAI

Enroll for Free



Anatomy of an Agent

- Trip planning
  - i.e. Book me a trip to San Francisco 
    - Figure out which tool to call
    - Search API
    - Use context
    - Construct a response
    - Overall correctness

Agent Evaluation Example

- Trip planning
  - i.e. Book me a trip to San Francisco 🌉
    - Figure out which tool to call (tool selection evaluation or eval)
    - Search API (function calling eval)
    - Use context (RAG eval)
    - Construct a response (tone eval)
    - Overall correctness (correctness eval)

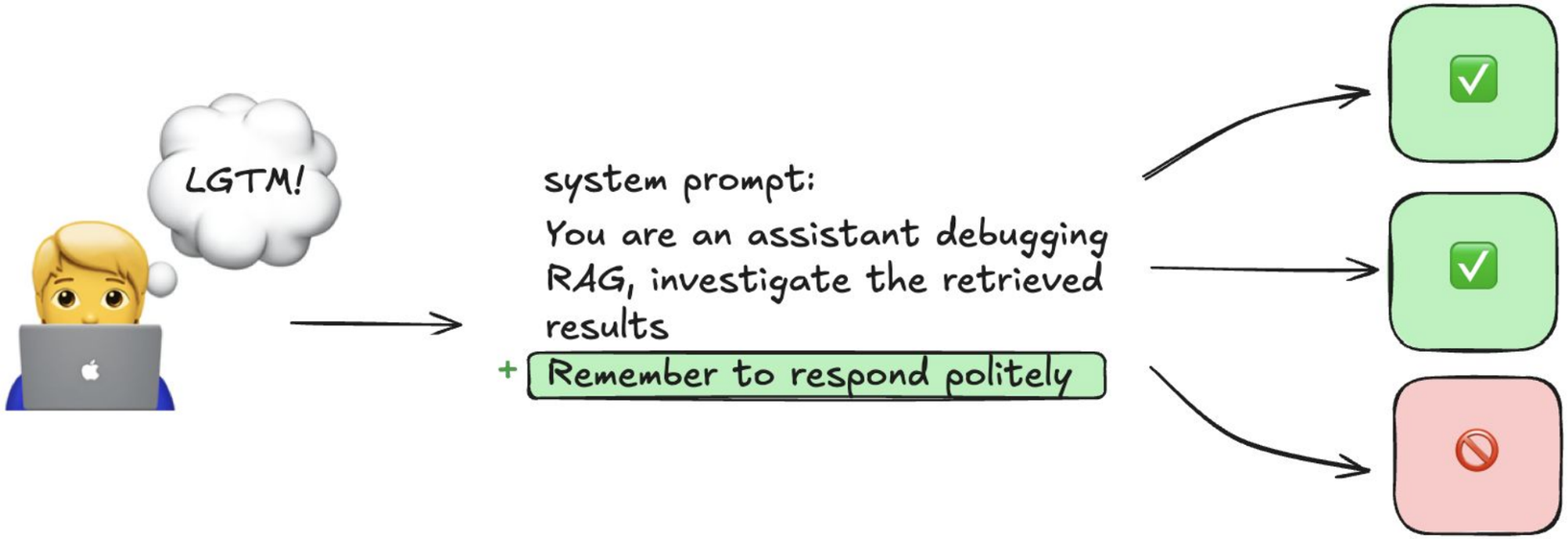
What could go wrong?

- Trip planning
  - i.e. Book me a trip to ~~San Francisco~~ San Diego 🏖️
    - ★ Figure out which tool to call (tool selection eval) → Calls the wrong tool
    - Search API (function calling eval) → Constructs search incorrectly
    - ★ Use context (RAG eval)
    - ★ Construct a response (tone eval) → Could be jailbroken, or inappropriate
    - Overall correctness (correctness eval) → Unhappy user ☹️

★ LLM-as-a-judge

Results in poor business outcomes and unhappy users!

Even Small Changes Can Cause Performance Regressions



Reality of AI Engineering:

Change a prompt or model, break a use case. Repeat.

Software Testing vs Agent Evals

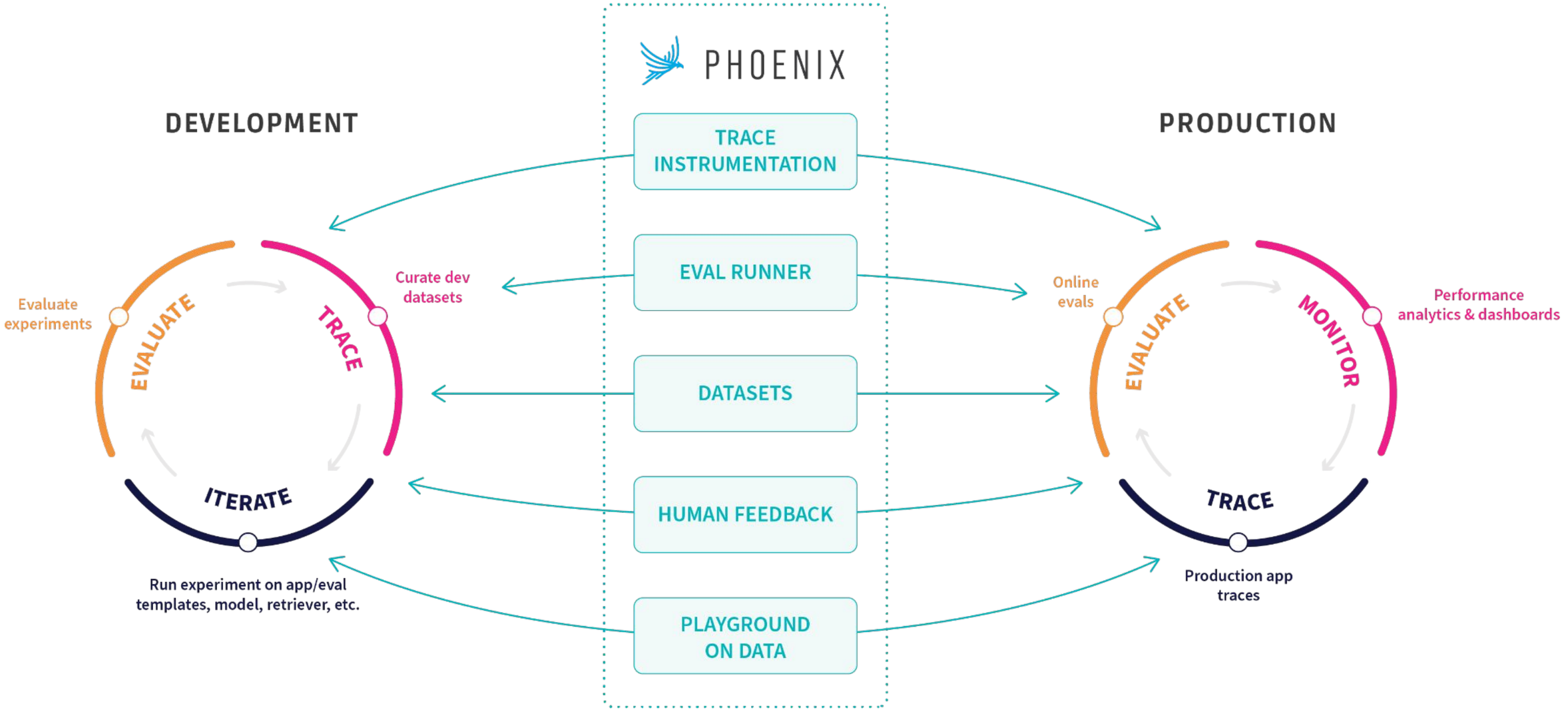
- Software is **deterministic**
- Unit tests are **deterministic**
- Integration tests rely on existing codebase and documentation

- LLM Agents are non **deterministic**
- LLM Agents can have **multiple paths**
- Improving agents relies on data

arize

DeepLearning.AI

Tools you will use to evaluate agents



What you've learned:

1. Scope of LLM evaluation
2. How AI Agents can be used to perform tasks
3. Similarities and differences between evaluating software and AI agents
4. Tools that are used to evaluate agents